

AERO40006 - Mathematics 1

Tutorial 1

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Question 1

Differentiate the following functions:

(1)	$f(x) = 5x^3 + x^{-1} - x^{-2}$	(2)	$f(x) = \sin(x) \cos(x)$	(3)	$f(x) = x \cos(x)$
(4)	$f(x) = x^2 \sin(x)$	(5)	$f(x) = \cos(x)/x$	(6)	$f(x) = e^x \cos(x)$
(7)	$f(x) = x^n \ln(x)$	(8)	$f(x) = \ln(n+x)$	(9)	$f(x) = \tan(x)$

$$(1) f'(x) = 15x^2 - x^{-2} + 2x^{-3}$$

$$(2) f'(x) = \cos^2(x) - \sin^2(x)$$

$$(3) f'(x) = \cos(x) - x \sin(x)$$

$$(4) f'(x) = 2x \sin(x) + x^2 \cos(x)$$

$$(5) f'(x) = \frac{-x \sin(x) - \cos(x)}{x^2}$$

$$(6) f'(x) = e^x \cos(x) - e^x \sin(x) = e^x (\cos(x) - \sin(x))$$

$$(7) f'(x) = n x^{n-1} \ln(x) + \frac{x^n}{x}$$

$$(8) f'(x) = \frac{1}{n+x}$$

$$(9) f'(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$f'(x) = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x)$$

Question 2

Calculate the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ from **first principles** when $f(x, y)$ is equal to:

(1)	x^2y	(2)	e^{xy}
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$$\begin{aligned} \text{(1)} \quad \frac{\partial f}{\partial x} &= 2xy & \frac{\partial f}{\partial x} &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2y - x^2y}{h} \\ & & &= \lim_{h \rightarrow 0} \frac{2hxy + h^2y}{h} = \lim_{h \rightarrow 0} (2xy + hy) = 2xy \\ \frac{\partial f}{\partial y} &= x^2 \end{aligned}$$

$$\frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{x^2(y+h) - x^2y}{h} = \lim_{h \rightarrow 0} x^2 = x^2$$

$$\begin{aligned} \text{(2)} \quad \frac{\partial f}{\partial x} &= ye^{xy} & \frac{\partial f}{\partial x} &= \lim_{h \rightarrow 0} \frac{e^{x+h,y} - e^{xy}}{h} = \lim_{h \rightarrow 0} \frac{e^{xy}e^{hy} - e^{xy}}{h} \\ & & &= e^{xy} \lim_{h \rightarrow 0} \frac{e^{hy} - 1}{h} = ye^{xy} \lim_{h \rightarrow 0} \frac{e^{hy} - 1}{hy} \\ & & &= ye^{xy} \lim_{h \rightarrow 0} \frac{e^{hy}}{1} = ye^{xy} \\ \frac{\partial f}{\partial y} &= xe^{xy} & \frac{\partial f}{\partial y} &= \lim_{h \rightarrow 0} \frac{e^{x(y+h)} - e^{xy}}{h} = xe^{xy} \end{aligned}$$

Question 3

Find the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ when $f(x, y)$ is equal to:

(1)	$\frac{x + 2y^2 + 1}{1 - y}$	(2)	x^y	(3)	$\arcsin\left(\frac{x}{y}\right)$	(4)	$\ln(x \ln(y))$
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$$(1) \quad \frac{\partial f}{\partial x} = \frac{(1-y) - 0}{(1-y)^2} = \frac{1}{1-y}$$

$$\frac{\partial f}{\partial y} = \frac{4y(1-y) + (x + 2y^2 + 1)}{(1-y)^2} = \frac{-2y^2 + 4y + x + 1}{y^2 - 2y + 1}$$

$$(2) \quad \frac{\partial f}{\partial x} = y x^{y-1}$$

$$f = x^y, \ln f = \ln x^y = y \ln x, (\ln f)' = \frac{1}{f} \cdot \frac{\partial f}{\partial y} = \ln x \quad \therefore \frac{\partial f}{\partial y} = f \ln x = x^y \ln x$$

$$(3) \quad f = \arcsin\left(\frac{x}{y}\right) \quad \therefore \frac{x}{y} = \sin f$$

$$\therefore \frac{1}{y} \frac{\partial x}{\partial f} = \cos f \quad \therefore \frac{\partial f}{\partial x} = \frac{1}{y \cos f} = \frac{1}{y \sqrt{1 - (\sin f)^2}} = \frac{1}{y \sqrt{1 - \frac{x^2}{y^2}}}$$

$$\therefore \frac{y}{x} = \frac{1}{\sin f}$$

$$\therefore \frac{1}{x} \frac{\partial y}{\partial f} = -\operatorname{cosec} f \cot f = \frac{-1}{\sin f \tan f} = \frac{\cos f}{-\sin^2 f} = \frac{\sqrt{1 - (\sin f)^2}}{-\sin^2 f} = \frac{\sqrt{1 - (\frac{x}{y})^2}}{-(\frac{x}{y})^2}$$

$$\therefore \frac{df}{dy} = \frac{-x}{y^2 \sqrt{1 - (\frac{x}{y})^2}}$$

$$(4) \quad u = x \ln y \quad \therefore \frac{du}{dx} = \ln y \quad \therefore \frac{du}{dy} = \frac{x}{y}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{du}{dx} = \frac{1}{x \ln y} \cdot \ln y = \frac{1}{x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{du}{dy} = \frac{1}{x \ln y} \cdot \frac{x}{y} = \frac{1}{y \ln y}$$

Question 4

Find $\frac{\partial^3}{\partial x^2 \partial y} (xe^y + x^3 y^4)$ and $\frac{\partial^4}{\partial x^2 \partial y^2} (\sin(xy))$.

$$\frac{\partial}{\partial x} = e^y + 3x^2 y^4$$

$$\frac{\partial}{\partial x} = y \cos(xy)$$

$$\frac{\partial^2}{\partial x^2} = 6xy^4$$

$$\frac{\partial^2}{\partial x^2} = -y^2 \sin(xy)$$

$$\frac{\partial^3}{\partial x^2 \partial y} = 24xy^3$$

$$\frac{\partial^3}{\partial x^2 \partial y} = -2y \sin(xy) - xy^2 \cos(xy)$$

$$\frac{\partial^4}{\partial x^2 \partial y^2} = -2 \sin(xy) - 2xy \cos(xy) - 2xy \cos(xy) + x^2 y^2 \sin(xy)$$

$$= (-2 + x^2 y^2) \sin(xy) - 4xy \cos(xy)$$

Question 5

Find the second derivatives of:

(1)	$f(x) = 1 + x^6 - x - x^{-2}$	(2)	$f(x) = x \cos(nx)$	(3)	$f(x) = e^x - e^{-x}$
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$$(1) f'(x) = 6x^5 - 1 + 2x^{-3}$$

$$f''(x) = 30x^4 - 6x^{-4}$$

$$(2) f'(x) = \cos(nx) - nx \sin(nx)$$

$$f''(x) = -n \sin(nx) - n \sin(nx) - n^2 x \cos(nx)$$

$$= -2n \sin(nx) - n^2 x \cos(nx)$$

$$(3) f'(x) = e^x + e^{-x}$$

$$f''(x) = e^x - e^{-x}$$

Question 6

Find $\frac{dy}{dx}$ in the following cases (simplify when necessary).

(1)	$y^2 = \cos(xy)$	(2)	$y = x^{\ln(x)}$	(3)	$5x - y + e^{xy} = 1$
(4)	$\ln(y) = x^2$	(5)	$\sqrt{y} = e^{x+x^2}$	(6)	$y = x \cosh(x)$

$$(1) \quad u = xy \quad \therefore \frac{du}{dx} = y \quad \therefore 2y \frac{dy}{dx} = -\sin(xy) \cdot y = -y \sin(xy) \quad \therefore \frac{dy}{dx} = \frac{-\sin(xy)}{2y + x \sin(xy)}$$

$$(2) \quad \ln y = \ln x^{\ln x} = \ln x \ln x \quad \therefore \frac{1}{y} \frac{dy}{dx} = \frac{2}{x} \ln x \quad \therefore \frac{dy}{dx} = \frac{2y \ln x}{x}$$

$$(2) \quad u = \ln(x) \quad \therefore \frac{du}{dx} = \frac{1}{x} \quad \therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = ux^{u-1} \cdot \frac{1}{x} = \ln(x) x^{\ln(x)-2}$$

$$(3) \quad 5 - \frac{dy}{dx} + ye^{xy} = 0 \quad \therefore \frac{dy}{dx} = 5 + ye^{xy} + xe^{xy} \frac{dy}{dx} \quad \therefore \frac{dy}{dx} = \frac{5 + ye^{xy}}{1 - xe^{xy}}$$

$$(4) \quad \frac{1}{y} \frac{dy}{dx} = 2x \quad \therefore \frac{dy}{dx} = 2xy$$

$$(5) \quad u = x + x^2 \quad \therefore \frac{du}{dx} = 1 + 2x$$

$$\frac{1}{2} y^{-\frac{1}{2}} \frac{dy}{dx} = \frac{1}{2y^{\frac{1}{2}}} \frac{dy}{dx} = (1+2x) e^{x+x^2} \quad \therefore \frac{dy}{dx} = 2\sqrt{y} (1+2x) e^{x+x^2}$$

$$(6) \quad y = x \left(\frac{e^x + e^{-x}}{2} \right) = \frac{xe^x}{2} + \frac{xe^{-x}}{2}$$

$$\frac{dy}{dx} = \frac{1}{2} e^x + \frac{1}{2} x e^x + \frac{1}{2} e^{-x} - \frac{1}{2} x e^{-x}$$

$$= \frac{1}{2} (1+x) e^x + \frac{1}{2} (1-x) e^{-x}$$

Question 7

Find a general formula for $\frac{d^n y}{dx^n}$ in each of the following cases:

(1)	$y = e^{2x}$	(2)	$y = \ln(1-x)$	(3)	$y = x^2 e^{2x}$
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$$(1) \frac{d^n y}{dx^n} = 2^n e^{2x}$$

$$(2) \frac{d^n y}{dx^n} = \frac{(-1)^n (n-1)!}{(1-x)^n} = \frac{(n-1)!}{(1-x)^n}$$

$$(3) \frac{d^n y}{dx^n} = \sum_{r=0}^n {}^n C_r (x^2)^{(n-r)} (e^{2x})^{(r)} = \sum_{r=0}^n {}^n C_r e^{2x} x^{2(n-r)}$$

$$= {}^n C_0 (x^2 e^{2x}) + {}^n C_1 (2x \cdot 2e^{2x}) + {}^n C_2 (2 \cdot 4e^{2x}) + {}^n C_3 (0 \cdot 8e^{2x}) + \dots$$

$$= x^2 e^{2x} + n(4xe^{2x}) + \frac{1}{2}n(n-1)(8e^{2x}) + 0 + \dots$$

$$= x^2 e^{2x} + 4nx e^{2x} + 4n^2 e^{2x} - 4ne^{2x}$$

$$= e^{2x} (x^2 + 4nx + 4n^2 - 4n)$$

$$= {}^n C_0 (e^{2x})^{(n)} x^2 + {}^n C_1 (e^{2x})^{(n-1)} \cdot 2x + {}^n C_2 (e^{2x})^{(n-2)} \cdot 2 + {}^n C_3 (e^{2x})^{(n-3)} \cdot 0$$

$$= 2^n e^{2x} x^2 + n \cdot (2^{n-1} e^{2x}) \cdot 2x + \frac{n(n-1)}{2} (2^{n-2} e^{2x}) \cdot 2$$

$$= (2^n x^2 + n2^n x + n(n-1)2^{n-2}) e^{2x}$$

Question 8

By using the Newton-Raphson procedure,

$$\frac{1}{3}x^3 - 3x^2 + 8x - 3 = 0$$

find the root which is close to $x = 0$ (correct to 3 decimal places).

$$f(x) = x^3 - 9x^2 + 24x - 9$$

$$x_0 = 0$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = -\frac{3}{8}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{93}{380}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.4244$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.4457$$

$$x_5 = 0.4459$$

$$\therefore x = 0.446$$

Question 9

Find the second order Taylor series approximations of

(i) $f(x, y) = \frac{1}{(1-x)y}$ about the point $(x, y) = (0, 1)$.

(ii) $f(x, y) = \frac{\cos(x)}{1 + \sin(y)}$ about the point $(x, y) = (0, 0)$.

$$(i) f_x = (1-x)^{-2}y^{-1} = \frac{1}{(1-x)^2y}$$

$$f_y = -(1-x)^{-1}y^{-2} = \frac{-1}{(1-x)y^2}$$

$$f_{xx} = 2(1-x)^{-3}y^{-1} = \frac{2}{(1-x)^3y}$$

$$f_{xy} = f_{yx} = -(1-x)^{-2}y^{-3} = \frac{-1}{(1-x)^2y^3}$$

$$f_{yy} = 2(1-x)^{-1}y^{-3} = \frac{2}{(1-x)y^3}$$

$$f(0, 1) + (x-0)f_x(0, 1) + (y-1)f_y(0, 1)$$

$$+ \frac{1}{2}[(x-0)^2 f_{xx}(0, 1) + 2(x-0)(y-1)f_{xy}(0, 1) + (y-1)^2 f_{yy}(0, 1)] + \dots$$

$$= 1 + x \cdot 1 + (y-1)(-1) + \frac{1}{2}[x^2 \cdot 2 + 2x(y-1)(-1) + (y-1)^2 \cdot 2] + \dots$$

$$= 1 + x - y + 1 + \frac{1}{2}(2x^2 - 2xy + 2x + 2y^2 - 4y + 2) + \dots$$

$$= 2 + x - y + x^2 - xy + x + y^2 - 2y + 1 + \dots$$

$$= 3 + 2x - 3y - xy + x^2 + y^2 + \dots$$

$$(ii) f_x = -\sin(x)(1 + \sin(y))^{-1} = \frac{-\sin(x)}{1 + \sin(y)}$$

$$f_y = -\cos(x)\cos(y)(1 + \sin(y))^{-2} = \frac{-\cos(x)\cos(y)}{(1 + \sin(y))^2}$$

$$f_{xx} = -\cos(x)(1 + \sin(y))^{-1} = \frac{-\cos(x)}{1 + \sin(y)}$$

$$f_{xy} = \sin(x)\cos(y)(1 + \sin(y))^{-2} = \frac{\sin(x)\cos(y)}{(1 + \sin(y))^2}$$

$$f_{yy} = \cos(x)\sin(x)(1 + \sin(y))^{-2} + 2\cos(x)\cos^2(y)(1 + \sin(y))^{-3} = \frac{\sin(x)\cos(x)}{(1 + \sin(y))^2} + \frac{2\cos(x)\cos^2(y)}{(1 + \sin(y))^3}$$

$$f(0, 0) + xf_x(0, 0) + yf_y(0, 0) + \frac{1}{2}[x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)]$$

$$= 1 + 0 + y(-1) + \frac{1}{2}(-x^2 + 0 + 2y^2)$$

$$= 1 - y - \frac{1}{2}x^2 + y^2$$

Question 10

Find the stationary points of

(i) $f(x, y) = x^4 - 2x^2 - y^2 + 2xy$

(ii) $f(x, y) = e^{-x^3+3x-y^2}$

and classify whether they are minima, maxima or saddle points.

(i) $f_x = 4x^3 - 4x + 2y, f_y = -2y + 2x$

$$\therefore f_x = f_y = 0 \quad \therefore 4x^3 - 4x + 2y = -2y + 2x = 0 \quad \therefore x = y \quad \therefore x(2x^2 - 1) = 0 \quad \therefore x = 0 \text{ or } \pm\sqrt{\frac{1}{2}}$$

$$\therefore \text{stationary points } (0, 0), \left(\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}\right), \left(-\sqrt{\frac{1}{2}}, -\sqrt{\frac{1}{2}}\right)$$

$$\Delta = (f_{xy})^2 - f_{xx}f_{yy} = (2)^2 - (12x^2 - 4)(-2) = 4 + 24x^2 - 8 = 24x^2 - 4$$

$$\Delta(0, 0) < 0, f_{yy}(0, 0) < 0, (0, 0) \text{ is a maximum}$$

$$\Delta\left(\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}\right) > 0, \left(\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}\right) \text{ is a saddle-point}$$

$$\Delta\left(-\sqrt{\frac{1}{2}}, -\sqrt{\frac{1}{2}}\right) > 0, \left(-\sqrt{\frac{1}{2}}, -\sqrt{\frac{1}{2}}\right) \text{ is a saddle-point}$$

(ii) $f_x = (-3x^2 + 3)e^{-x^3+3x-y^2}, f_y = -2ye^{-x^3+3x-y^2}$

$$f_x = f_y = 0 \quad \therefore (-3x^2 + 3) = -2y = 0 \quad \therefore y = 0 \quad \therefore x = \pm 1$$

$$\therefore \text{stationary points } (1, 0), (-1, 0)$$

$$f_{xx} = -6xe^{-x^3+3x-y^2} + (-3x^2+3)e^{-x^3+3x-y^2}$$

$$f_{yy} = -2e^{-x^3+3x-y^2} + 4y^2e^{-x^3+3x-y^2} = (4y^2 - 2)e^{-x^3+3x-y^2}$$

$$f_{xy} = (6x^2y - 6y)e^{-x^3+3x-y^2}$$

$$\Delta = (f_{xy})^2 - f_{xx}f_{yy}$$

$$\Delta(1, 0) = (6e^2)^2 + 6e^2 \times (-2)e^2 \stackrel{f_{xx} < 0}{< 0}, (1, 0) \text{ is a } \text{maximum point}$$

$$\Delta(-1, 0) = (6e^2)^2 - 6e^2 \times (-2)e^2 > 0, (-1, 0) \text{ is a saddle point}$$

Question 11

Find and classify the stationary points of the function $f(x, y) = x^n + y^n$ in the cases $n = 2, 3$ and 4.

$$f_x = nx^{n-1}, f_y = ny^{n-1}$$

$$f_x = f_y = 0 \quad \therefore nx^{n-1} = ny^{n-1} = 0 \quad \therefore \text{stational point } (0, 0)$$

$$f_{xx} = n(n-1)x^{n-2}$$

$$f_{yy} = n(n-1)y^{n-2}$$

$$f_{xy} = 0$$

For $n = 2$,

$$\Delta(0, 0) = 0^2 - (2 \times 1)(2 \times 1) = -4 < 0, f_{xx} > 0, (0, 0) \text{ is a minimum}$$

For $n = 3$, (4)

$$f(0, 0) = 0, \text{ for any } h > 0, f(h, 0) > 0 \text{ and } f(-h, 0) < 0,$$

$$\Delta(0, 0) = 0, \lambda(0, 0) \text{ is a saddle-point}$$

For $n = 4$

$$\Delta(0, 0) = 0, f(0, 0) = 0 \text{ and } f(x, y) > 0 \text{ (} x \neq 0, y \neq 0 \text{)}, (0, 0) \text{ is a minimum point}$$

Question 12

- (i) Find the stationary points of the function

$$f(x, y) = x^2 y^2 - \left(x + y - xy - \frac{1}{2} \right)^2.$$

- (ii) Now suppose that a triangle with side lengths
- x, y
- and
- z
- has perimeter of length 1. If the lengths
- x, y
- are such that a triangle of this form can exist, then its area
- A
- is related to
- $f(x, y)$
- by

$$4A^2 = f(x, y).$$

Discuss the meaning of the stationary points in this context.

$$i) f_x = 2xy^2 - 2(1-y)(x+y-xy-\frac{1}{2}) = 2xy^2 - 2(x+y-xy-\frac{1}{2} - xy - y^2 + xy^2 + \frac{1}{2}y)$$

$$= 2xy^2 - 2x - 3y + 4xy + 1 + 2y^2 - 2xy^2 = 2y^2 + 4xy - 2x - 3y + 1$$

$$f_y = 2x^2 y - 2(1-x)(x+y-xy-\frac{1}{2}) = 2x^2 y + 4xy - 2y - 3x + 1$$

$$f_x = f_y = 0 \quad \therefore 2y^2 + 4xy - 2x - 3y + 1 = 2x^2 + 4xy - 2y - 3x + 1 = 0 \quad \therefore 2y^2 - 2x - 3y = 2x^2 - 2y - 3x \quad \therefore 2y^2 - y - 2x^2 + x = 0$$

$$\therefore (y - \frac{1}{4})^2 - (x - \frac{3}{4})^2 = 0 \quad \therefore x = y \quad \text{or} \quad y = \frac{1}{2} - x$$

$$x_1 = \frac{1}{2}, y_1 = \frac{1}{2} \\ x_2 = \frac{1}{3}, y_2 = \frac{1}{3}$$

$$\text{For } x = y, 2x^2 + 4x^2 - 2x - 3x + 1 = 6x^2 - 5x + 1 = 0 \quad \therefore \text{no solution}$$

$$\text{For } y = \frac{1}{2} - x, 2x^2 + 4x(\frac{1}{2} - x) - 2(\frac{1}{2} - x) - 3x + 1 = 0 \quad \therefore x_1 = \frac{15 + \sqrt{193}}{4}, y_1 = -\frac{5 + \sqrt{193}}{4} \\ x_2 = \frac{15 - \sqrt{193}}{4}, y_2 = \frac{-5 + \sqrt{193}}{4}$$

$$2x^2 + 2x - 4x^2 - 1 + 2x - 3x + 1$$

$$= -2x^2 + x = 0$$

$$\therefore x_3 = \frac{1}{2}, y_3 = 0$$

$$x_4 = 0, y_4 = \frac{1}{2}$$

$$ii) f_{xy} = 4y + 4x - 3$$

$$f_{xx} = 4y - 2$$

$$f_{yy} = 4x - 2$$

$$\Delta(\frac{1}{2}, \frac{1}{2}) = 1^2 - 0 > 0, \text{ saddle point}$$

$$\Delta(\frac{1}{3}, \frac{1}{3}) = (-\frac{1}{3})^2 - (-\frac{2}{3})^2 = -\frac{1}{3} < 0, f_{xx} = f_{yy} = -\frac{2}{3} < 0, \text{ max point}$$

$$\Delta(\frac{1}{2}, 0) = (-1)^2 - (-2)(0) = 1 > 0, \text{ saddle point}$$

$$\Delta(0, \frac{1}{2}) = (-1)^2 - (0)(-2) = 1 > 0, \text{ saddle point}$$

$$\Delta(\frac{15 - \sqrt{193}}{4}, \frac{-5 + \sqrt{193}}{4}) = 7^2 - (-7 + \sqrt{193})(13 - \sqrt{193}) > 0$$

At the stationary point $(\frac{15 - \sqrt{193}}{4}, \frac{-5 + \sqrt{193}}{4})$, the area of the triangle will be at the saddle point

Question 13

Suppose $f(x, y)$ is a function of two variables. Let

$$x(r, \theta) = r \cos \theta, \quad y(r, \theta) = r \sin \theta$$

be polar co-ordinates and consider the function $u(r, \theta) = f(x(r, \theta), y(r, \theta))$. Show that if u satisfies the equation

$$(\cos \theta + \sin \theta) \frac{\partial u}{\partial r} + \frac{1}{r} (\cos \theta - \sin \theta) \frac{\partial u}{\partial \theta} = 0$$

then the original function f must satisfy

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0.$$

$$\frac{\partial u}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial f}{\partial x} + \sin \theta \frac{\partial f}{\partial y}$$

$$\frac{\partial u}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = -r \sin \theta \frac{\partial f}{\partial x} + r \cos \theta \frac{\partial f}{\partial y} = r (\cos \theta \frac{\partial f}{\partial y} - \sin \theta \frac{\partial f}{\partial x})$$

$$\text{If } (\cos \theta + \sin \theta) \frac{\partial u}{\partial r} + \frac{1}{r} (\cos \theta - \sin \theta) \frac{\partial u}{\partial \theta} = 0$$

$$\therefore (\cos \theta + \sin \theta) (\cos \theta \frac{\partial f}{\partial x} + \sin \theta \frac{\partial f}{\partial y}) + \frac{1}{r} (\cos \theta - \sin \theta) r (\cos \theta \frac{\partial f}{\partial y} - \sin \theta \frac{\partial f}{\partial x})$$

$$= \cos^2 \theta \frac{\partial f}{\partial x} + \cos \theta \sin \theta \frac{\partial f}{\partial x} + \cos \theta \sin \theta \frac{\partial f}{\partial y} + \sin^2 \theta \frac{\partial f}{\partial y} + \cos^2 \theta \frac{\partial f}{\partial y} - \cos \theta \sin \theta \frac{\partial f}{\partial x} - \cos \theta \sin \theta \frac{\partial f}{\partial y} + \sin^2 \theta \frac{\partial f}{\partial x}$$

$$= (\cos^2 \theta + \sin^2 \theta) \frac{\partial f}{\partial x} + (\sin^2 \theta + \cos^2 \theta) \frac{\partial f}{\partial y}$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$